CompSci 390-03-26S

Distributed and Networked Systems

The Internet

Channel abstraction

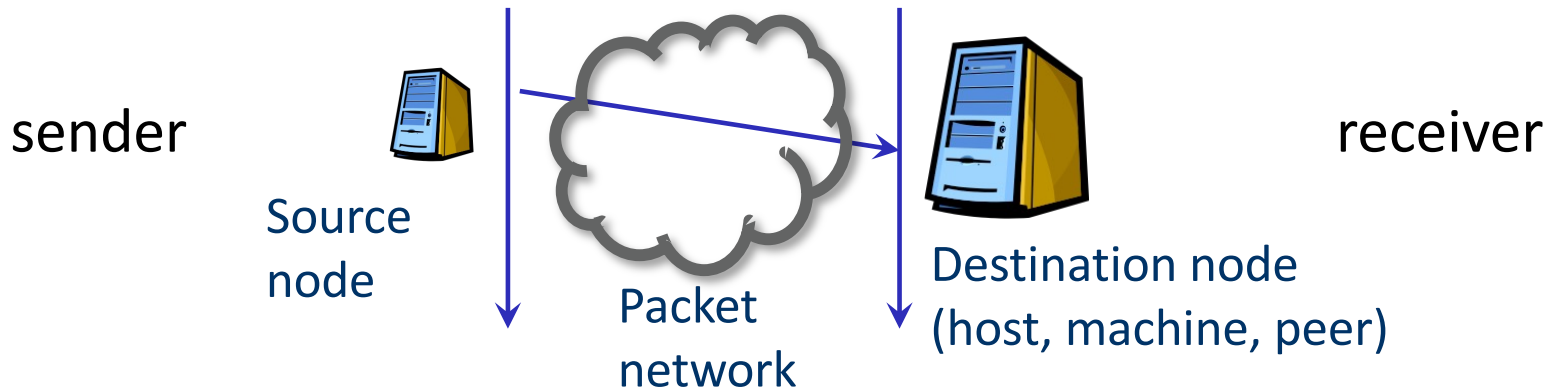
Distributed systems are made up of processes that send and receive messages over channels.

What's a **channel?**

- Transport for ordered stream of messages (or bytes)
- Connection between 2+ endpoints
- Notify S or R on close by the other
- In Go: many possible senders and receivers
- Internet connections: point-to-point
- How to support stream/channel abstraction over Internet?

Network communication

- Computers (nodes, hosts) send and receive **packets**.
- A packet is a block of bytes (**octets**: 8 bits), with some maximum size
  - Maximum Transmission Unit (**MTU**) imposed by networks on the packet's **path**
- A **message** may be sent as multiple packets (some reassembly required).
- Nodes often communicate with **streams** (sequences) of multiple packets over a **connection** or **session**, resulting in a **flow** of packets.



Packet networks: origins

THE INTERNET: PAST, PRESENT, AND FUTURE

The Beginnings of Packet Switching: Some Underlying Concepts

Paul Baran, Com21, Inc.

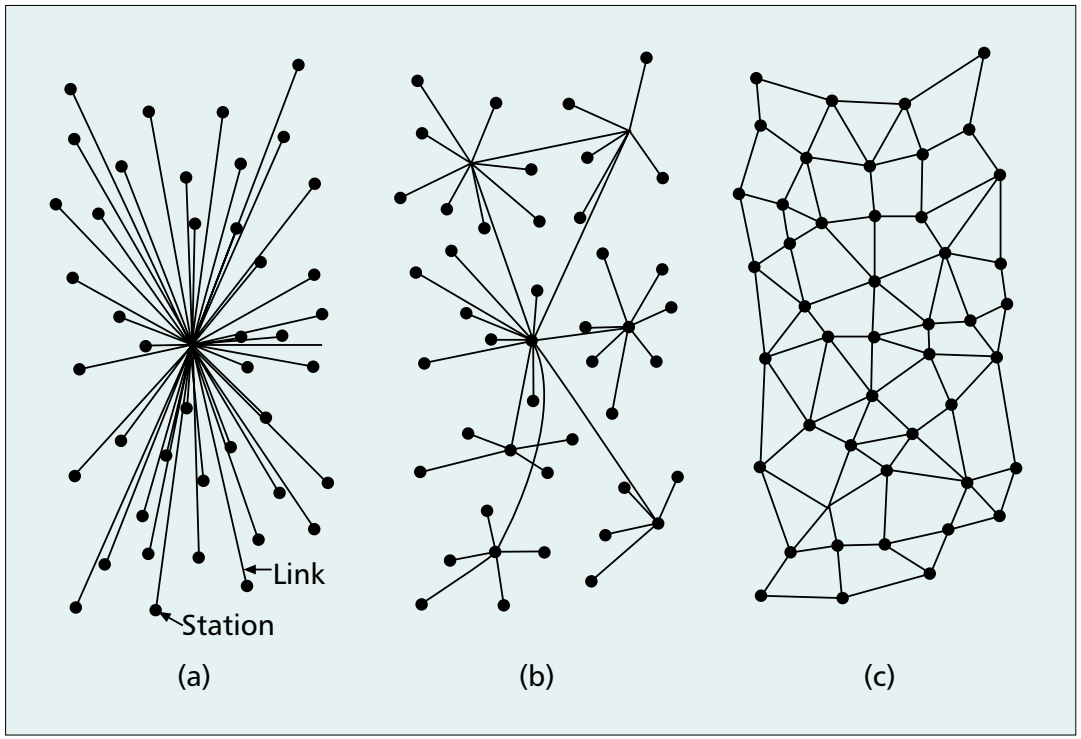
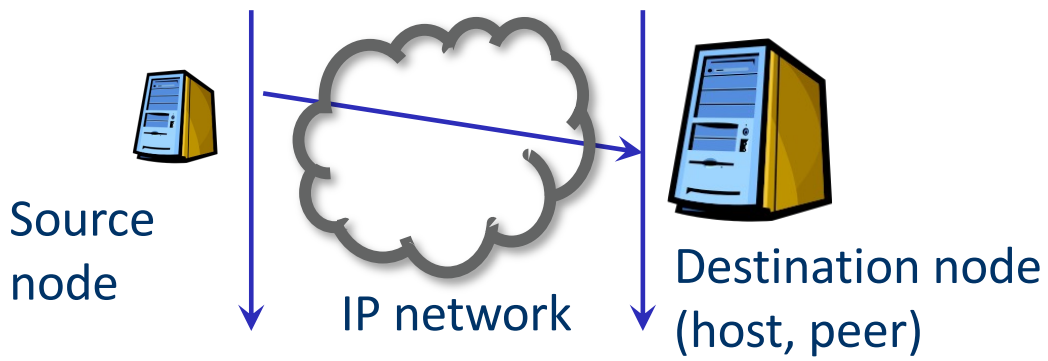


Figure 1. *Three forms of networks: a) centralized; b) decentralized; c) distributed.*

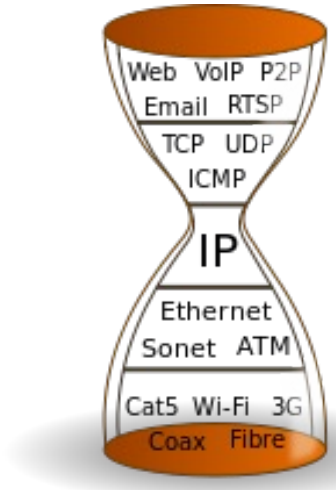
Internet communication

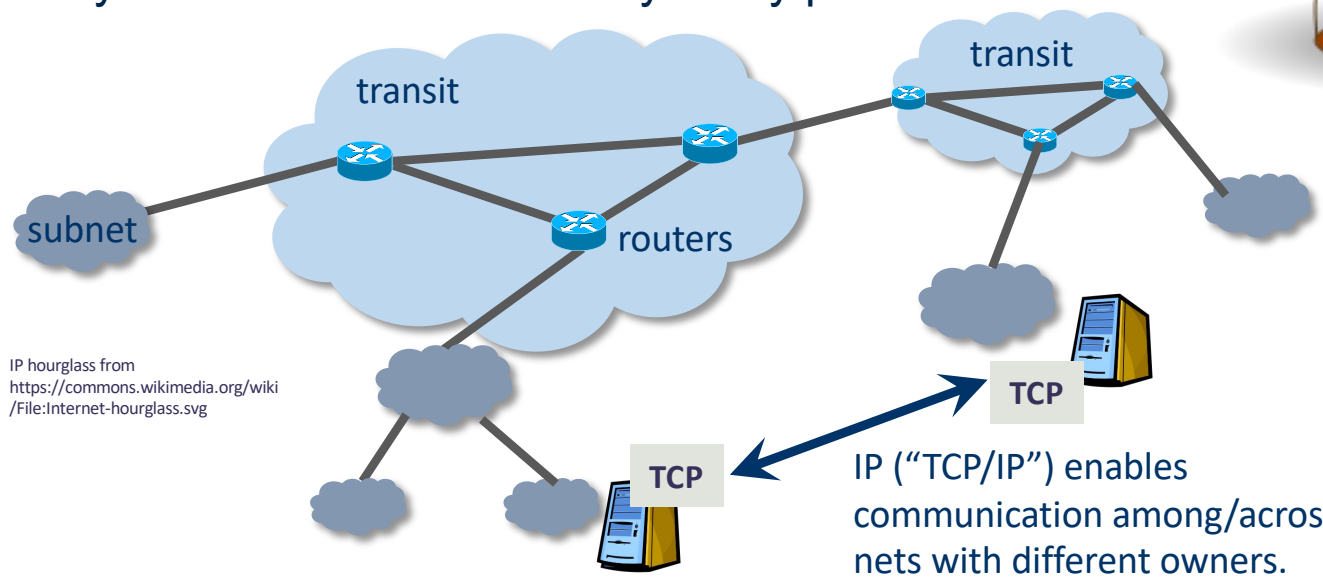
- We consider only networks using Internet Protocol (**IP**).
- Internet standards define the formats and rules (**protocols**) for communication over "networks of networks".
- It's all in the IETF "RFC" docs: thousands of them.
  - https://www.ietf.org/standards/rfcs/



The Internet

- ”Everything over IP and IP over everything.”
- Each Internet-connected computer (**node** or **host**) attaches to an **edge network** (a local subnet).
- A path may cross networks owned by many providers.



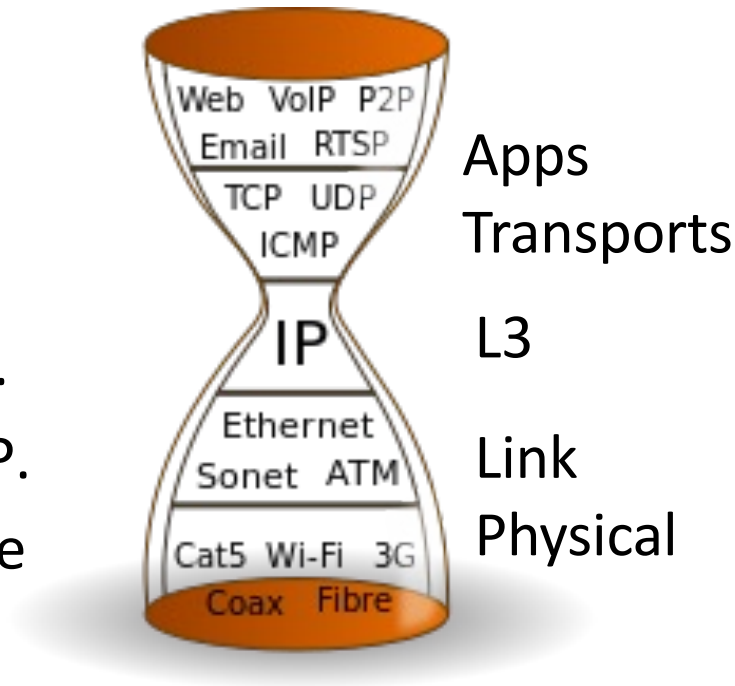


IP hourglass from https://commons.wikimedia.org/wiki/File:Internet-hourglass.svg

IP (“TCP/IP”) enables communication among/across nets with different owners.

IP: The thin waist of the hourglass

- Want to send a message?
- Wrap (**encapsulate**) it in an IP packet
- Put an **address** on it.
- Put a return address on it.
- In a **header** at the front of the packet.
- Header format defined by protocol: IP.
- Add another header/wrap to cross the local network (LAN, e.g., Ethernet)
- Send and wait.

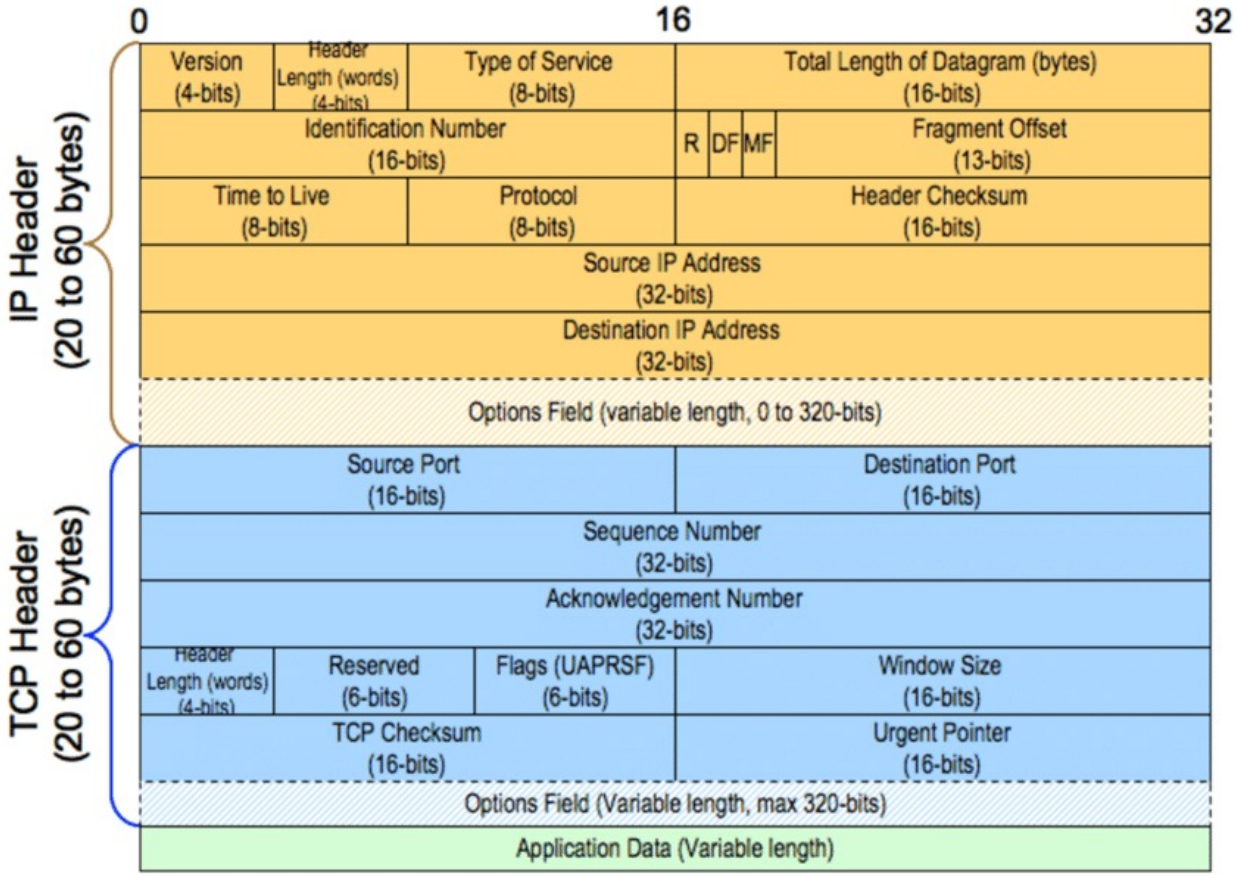


Packets

- A packet is a struct in memory on sender.
- Network software builds it.
- OS transmits it via network device using DMA.
- Bit values pass through the network path.
- DMA leaves a copy of the struct in receiver memory.
- Network software unwraps/delivers it.

When is it safe for sender to garbage-collect its copy of the packet?

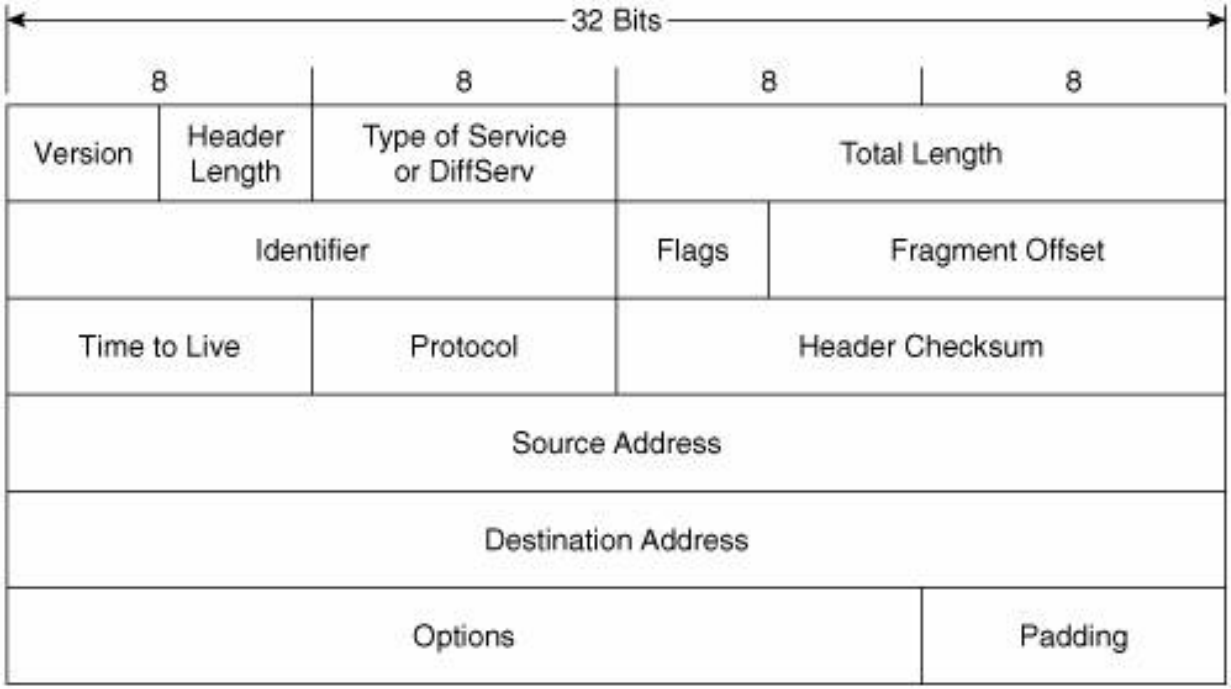
Packet headers on the Internet (e.g., TCP/IP)



Not on the test.

<u>TCP/IP protocol</u>
Data fields of various bit widths at specific offsets. Network byte order: **Big Endian**. May require **byte swapping**.

IPv4 header

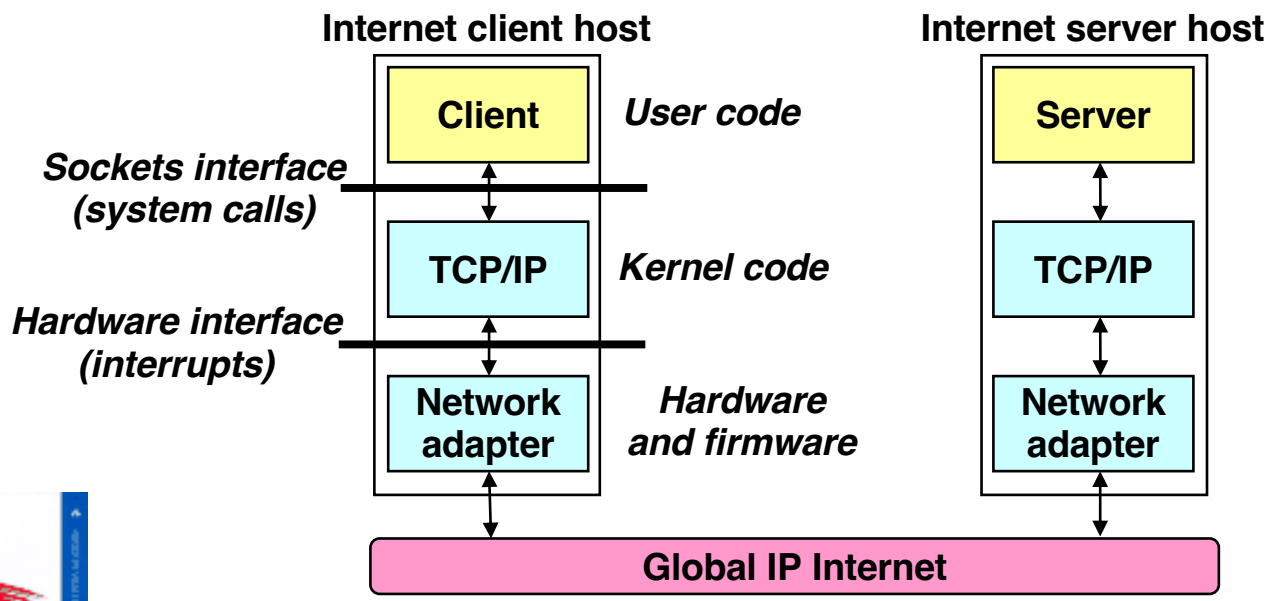


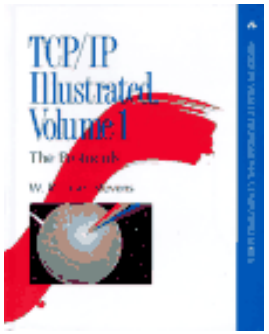
20 bytes

Born to be free and dumb

- **Dumb**: all the "smart" is at the edge: software on end nodes
  - **End-to-end principle**
- **Free**: open interconnection
  - Link and integrate many different technologies
  - Peering/attachment by mutual consent
- What does it mean at IP layer (L3)?
  - Routers just hand the packet off to the next hop on path
  - **Best effort**. Too much traffic? Just drop excess packets.
  - Channel properties implemented in transport on end nodes, e.g., TCP

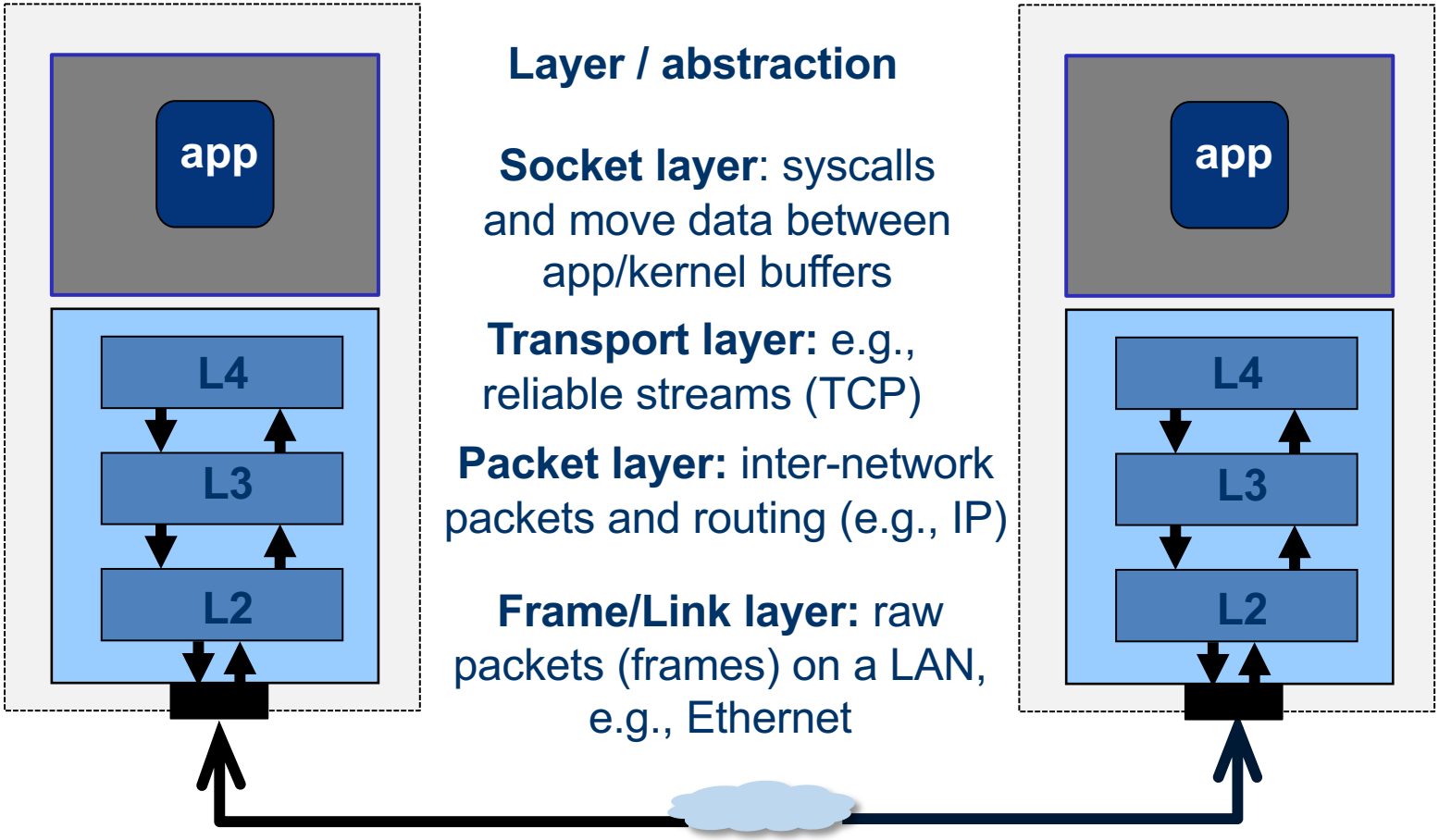
The network stack, simplified





The **network protocol stack** is a hierarchy/layering of software modules that implement **network protocols**: standard formats and rules for communicating with peers over a network.

Network protocol stack



Layer / abstraction

Socket layer: syscalls and move data between app/kernel buffers

Transport layer: e.g., reliable streams (TCP)

Packet layer: inter-network packets and routing (e.g., IP)

Frame/Link layer: raw packets (frames) on a LAN, e.g., Ethernet

The role of the transport (L4), e.g., TCP

- Negotiate/establish connection between two endpoints
- Reassemble packets into app-layer messages
- Reliable streams: sequences of bytes or messages
- What if a packet is dropped in transit?
- How to know it was dropped?
- Or packets arrive out of order?
- Flow control—if sender is “too fast”?

TRANSMISSION CONTROL PROTOCOL

DARPA INTERNET PROGRAM

PROTOCOL SPECIFICATION

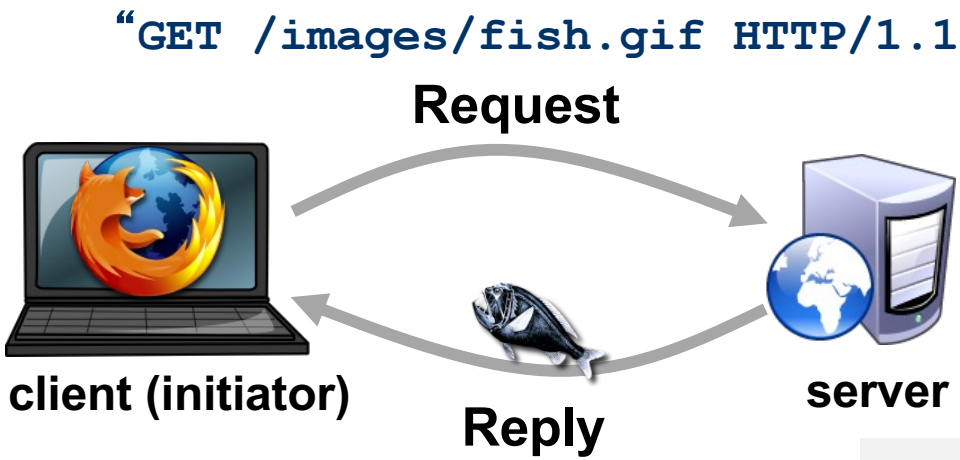
September 1981

RFC 793

TCP/IP Ports

- Each IP transport endpoint on a host has a logical port number (16 bits) that is unique on that host.
- This port abstraction is an Internet Protocol concept.
  - Source/dest port is named in every IP packet.
  - Kernel looks at port to demultiplex incoming traffic.
- What port number to connect to?
  - We have to agree on well-known ports for common services
  - E.g., 80 for HTTP/Web.
  - Network elements (e.g., firewall) may block traffic by port#.
- Clients pick a random local port # for response data.

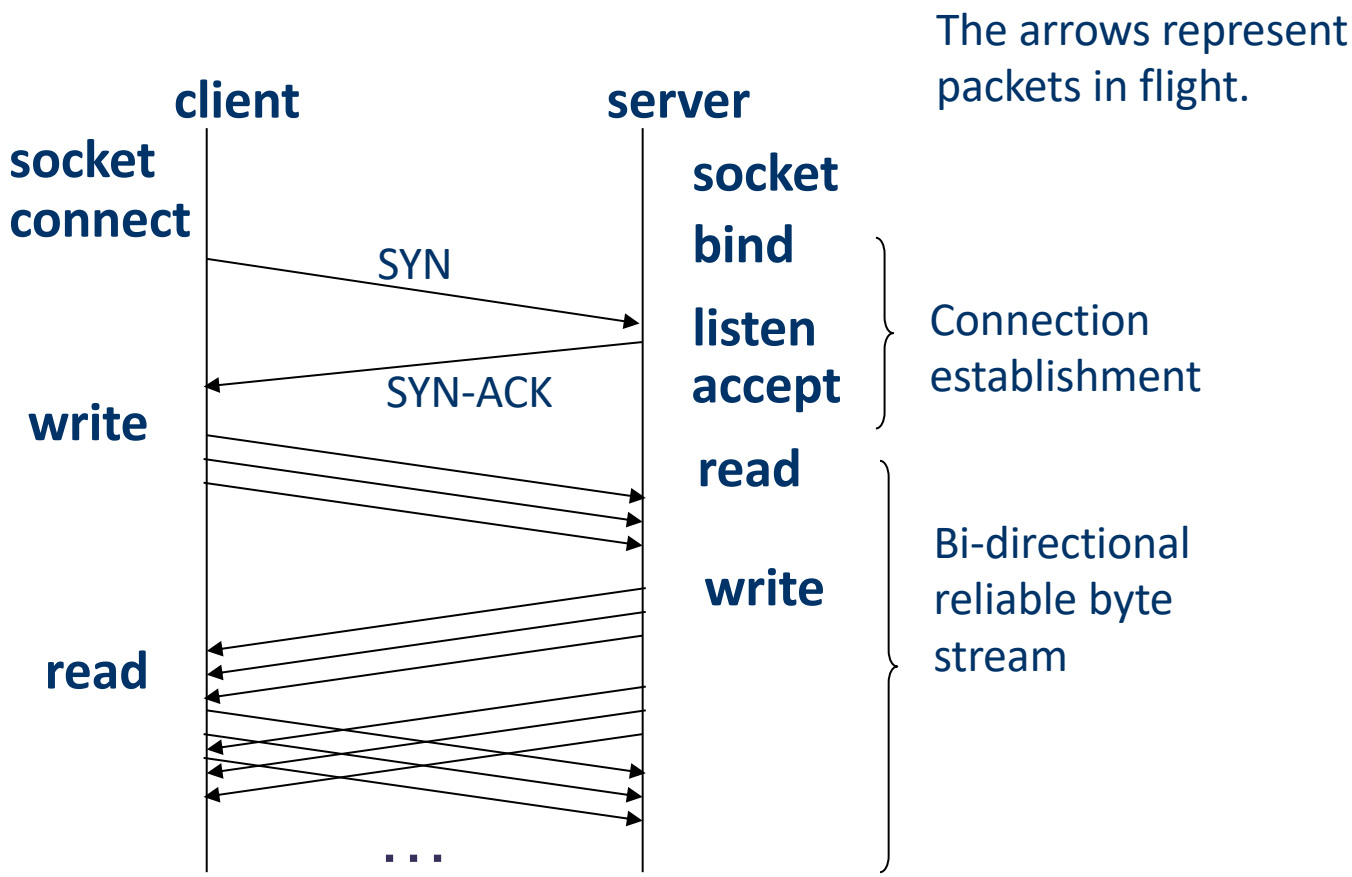
Clients, servers, and the socket API [L02]




```
sd = socket(…);
connect(sd, name+port);
write(sd, request…);
read(sd, reply…);
close(sd);
```

```
s = socket(…);
bind(s, name+port);
listen(s, N);
sd = accept(s);
read(sd, request…);
write(sd, reply…);
close(sd);
```

Stream over a TCP connection



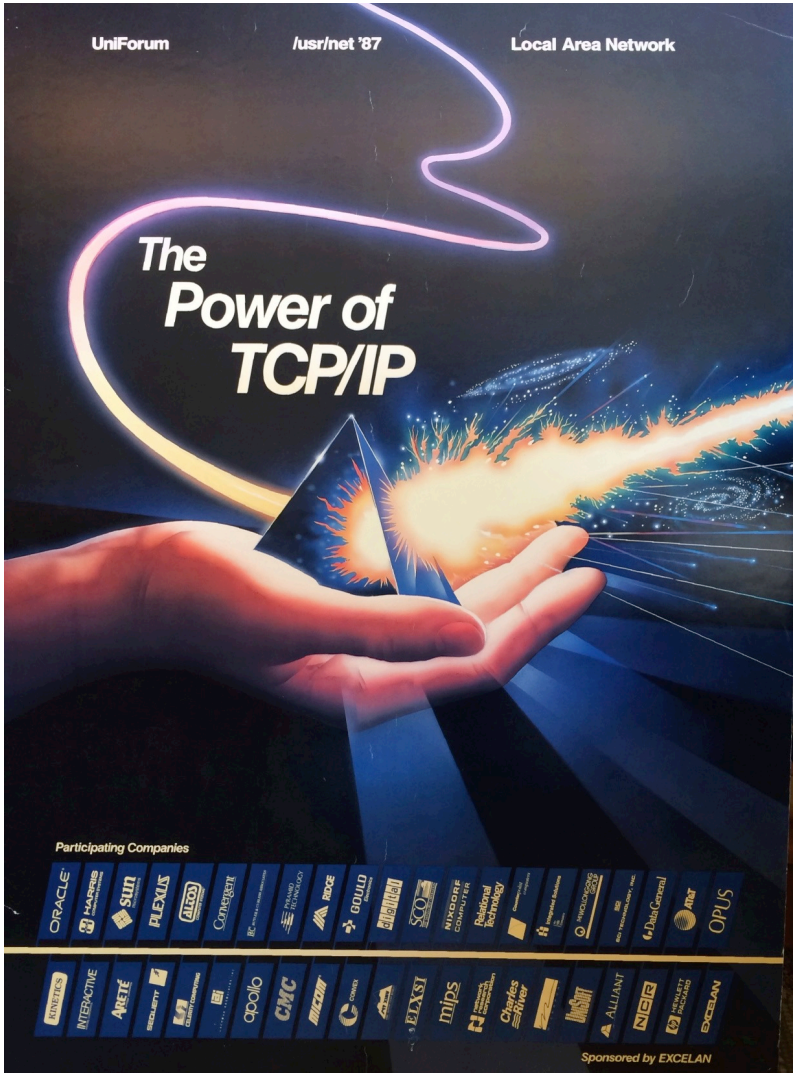
What is in the stream for a Web request? (HTTP 1.1)

```
GET /hello.txt HTTP/1.1
User-Agent: curl/7.63.0 libcurl/7.63.0
Host: www.example.com
Accept-Language: en
```

```
HTTP/1.1 200 OK
Date: Wed, 30 Jan 2019 12:14:39 GMT
Server: Apache
Last-Modified: Mon, 28 Jan 2019 11:17:01 GMT
Accept-Ranges: bytes
Content-Length: 12
Vary: Accept-Encoding
Content-Type: text/plain

Hello World!
```

https://www.cloudflare.com/learning/ssl/why-is-http-not-secure/



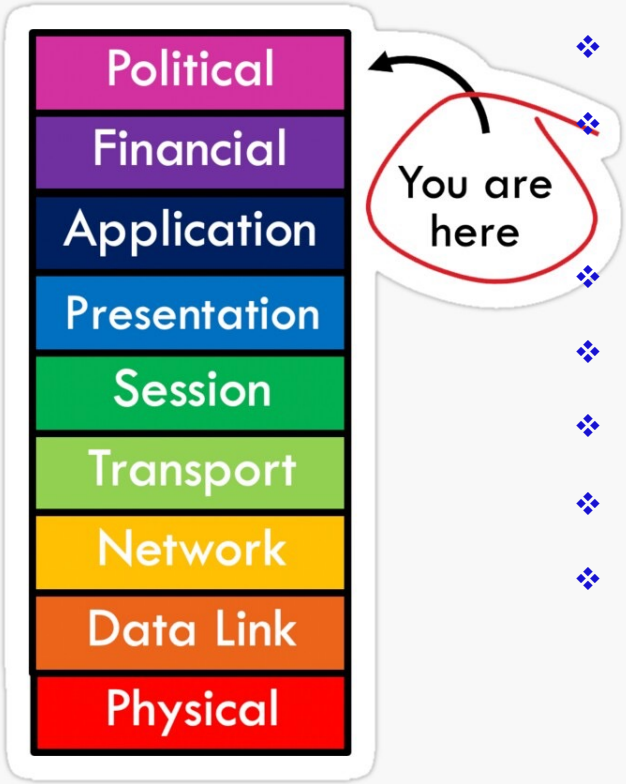
The Internet concept wasn't obviously compelling, at least not to everyone.

It had to be marketed, even within the tech community.

In 1986, the US National Science Foundation (NSF) opened the door to a commercial Internet (then NSFNET). IP support in sockets (Berkeley Unix) was widely used among academics.

The driving force for adopting TCP/IP was a collection of Unix-oriented startups and upstarts arrayed against a few large companies with their own proprietary network standards.

The Internet as a Grand Human Experiment



- In decentralized and resilient global infrastructure
- In engineering design
  - Strong foundations "push innovation upwards"
- In technical evolution guided by foundational principles
- In governance by institutions—non-profit NGOs
- In governance of institutions
- Governance by multi-stakeholder consensus
- In global cooperation for a global public good

60 Years So Far

IETF RFC 7282 (2014): On Consensus and Humming in the IETF

The IETF has had a long tradition of doing its technical work through a consensus process, taking into account the different views among IETF participants and coming to (at least rough) consensus on technical matters. In particular, the IETF is supposed not to be run by a "majority rule" philosophy. This is why we engage in rituals like "humming" instead of voting. However, more and more of our actions are now indistinguishable from voting, and quite often we are

Almost every IETF participant knows the aphorism from Dave Clark's 1992 plenary presentation [Clark] regarding how we make decisions in the IETF:

> We reject: kings, presidents and voting.
>
> We believe in: rough consensus and running code.

That is, our credo is that we don't let a single individual dictate decisions (a king or president), nor should decisions be made by a vote, nor do we want decisions to be made in a vacuum without practical experience. Instead, we strive to make our decisions by the consent of all participants, though allowing for some dissent (rough consensus), and to have the actual products of engineering (running code) trump theoretical designs.

Critical Properties of "The Internet Way of Networking"

- An Accessible Infrastructure with a **Common Protocol**
- An **Open Architecture** of Interoperable and Reusable Building Blocks
- **Decentralized Management** and a Single Distributed Routing System
- Common **Global Identifiers**
- A Technology Neutral, **General-Purpose** Network

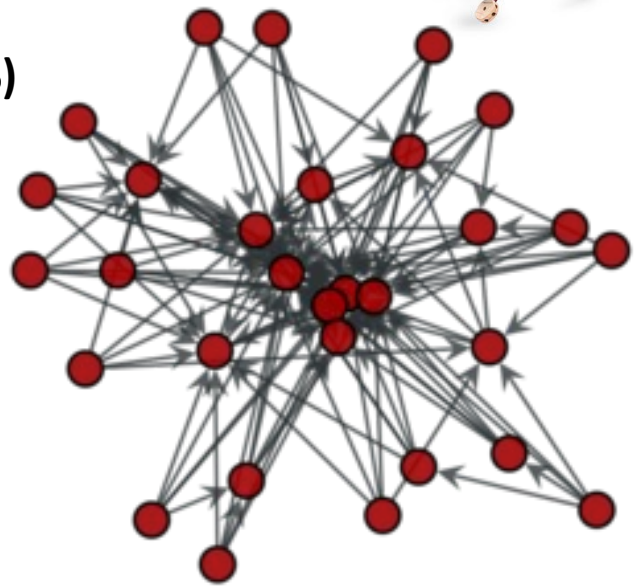
Internet Society

Critical Property	Benefits
1. An Accessible Infrastructure with a Common Protocol that is open and has low barriers to entry	Unrestricted access and common protocols deliver global connectivity and encourage the network to grow. As more and more participants connect, the value of the Internet increases for everyone.
2. Open Architecture of Interoperable and Reusable Building Blocks based on open standards development processes voluntarily adopted by a user community	Open architecture creates common interoperable services, which deliver fast and permissionless innovation everywhere. The inclusive standardization process and demand-driven adoption ensures that useful changes are adopted, while unnecessary ones disappear.
3. Decentralized Management and a Single Distributed Routing System which is scalable and agile	Distributed routing delivers a resilient and adaptable network of autonomous networks, allowing for local optimizations while maintaining worldwide connectivity.
4. Common Global Identifiers which are unambiguous and universal	A common identifier set delivers consistent addressability and a coherent view of the entire network, without fragmentation or fractures.
5. A Technology Neutral, General-Purpose Network which is simple and adaptable	Generality delivers flexibility. The Internet continuously serves a diverse and constantly evolving community of users and applications. It does not require significant changes to support this dynamic environment.

https://www.internetsociety.org/resources/internet-impact-assessment-toolkit/the-internets-five-critical-properties/

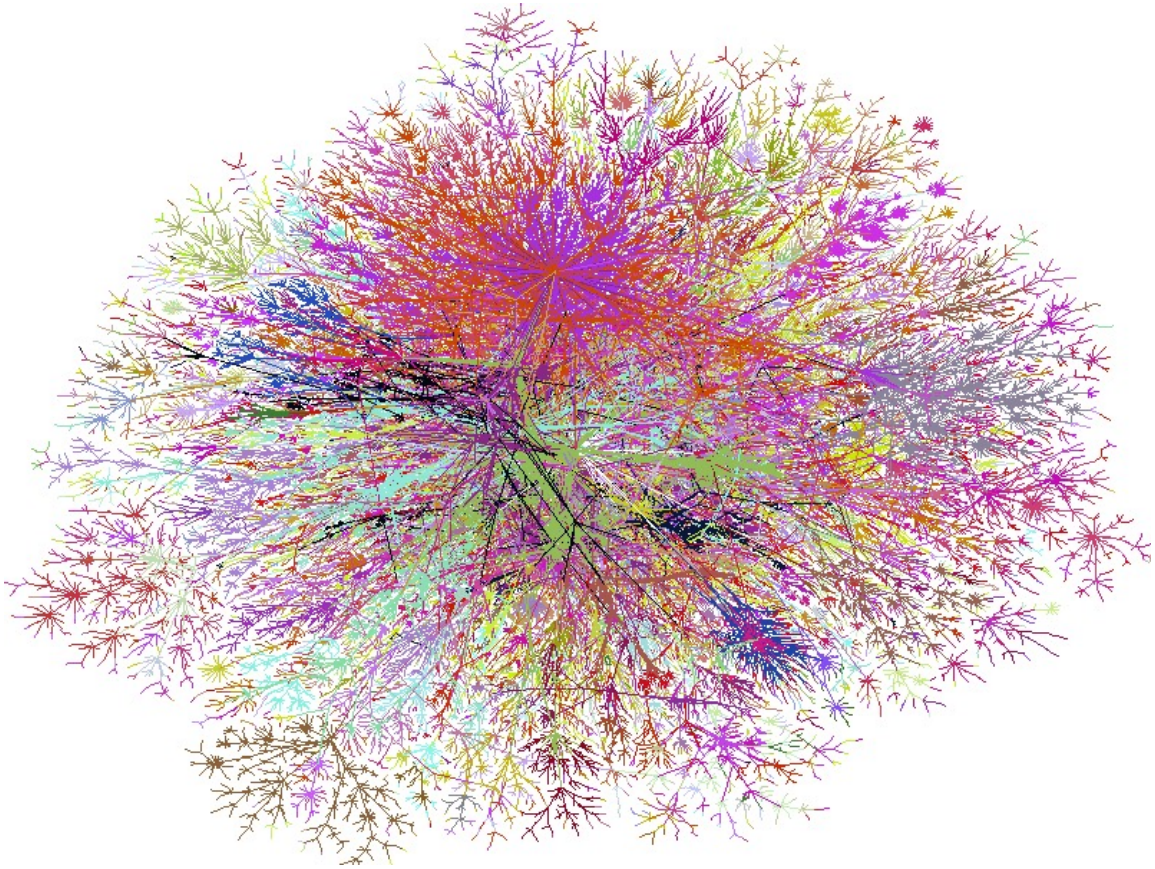
Physical topology and implications

- Open interconnection → network peering and topology are choices in an **economic** "game".
- Decentralized: independent actors respond to cost/benefit incentives, no "design".
  - Routing domains are **autonomous systems (AS)**
- Constantly evolving
- **Preferential attachment**
- "Choke points"?



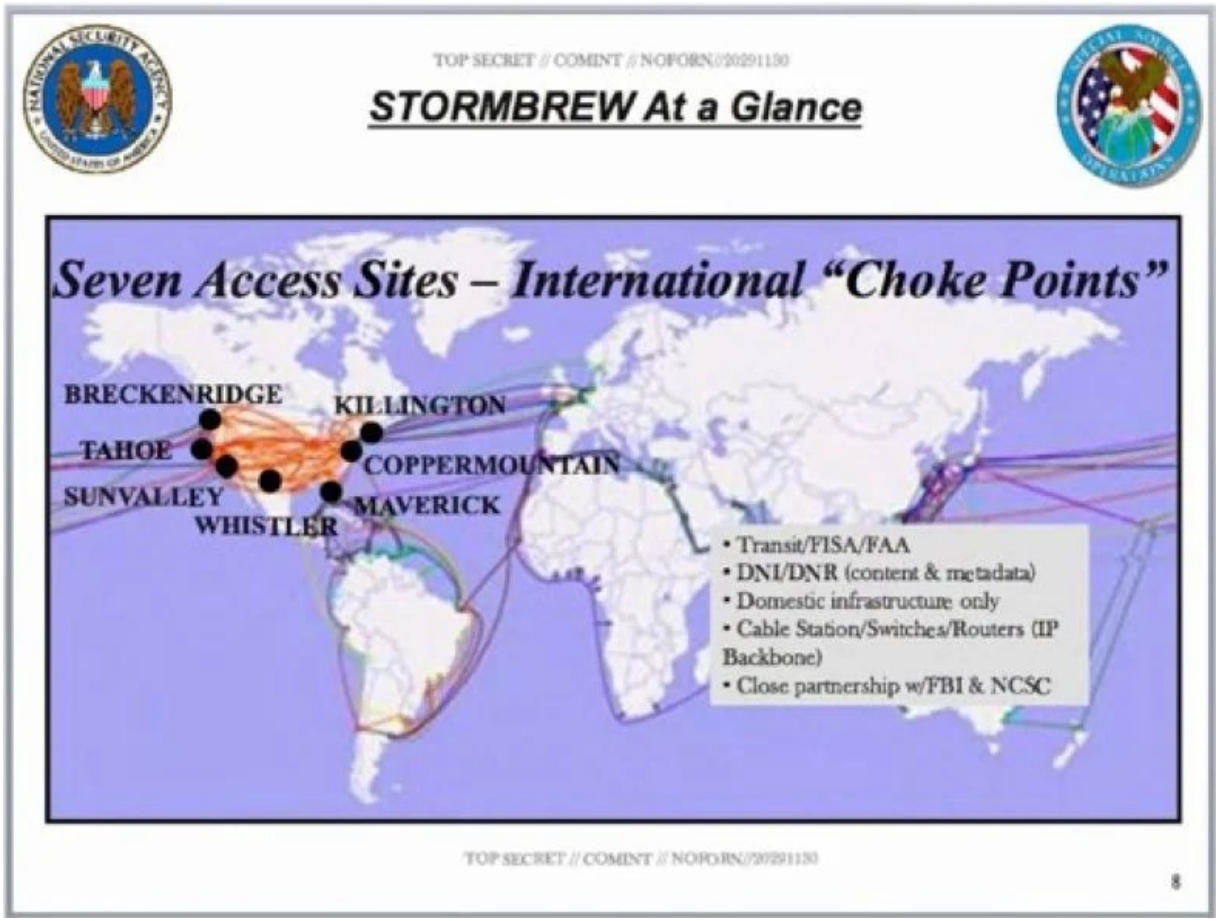
https://en.wikipedia.org/wiki/Preferential_attachment

December 1998, from Wired Magazine



The Internet Mapping Project, Bill Cheswick and Hal Burch, Lucent/Bell Labs, then Lumeta

Stellar Wind and Stormbrew



Path security

Anyone who owns a network on the transmit path can:

- intercept the packet
- transform/corrupt the bits
- drop the packet or misdirect

"Internet treats censorship as failure and routes around it."

Who governs the Internet?

ICANN

One World, One Internet

ICANN: an “internationally organized” nonprofit that is responsible for the coordination of…unique identifiers related to the namespaces of the Internet, and ensuring the network's stable and secure operation.

iana

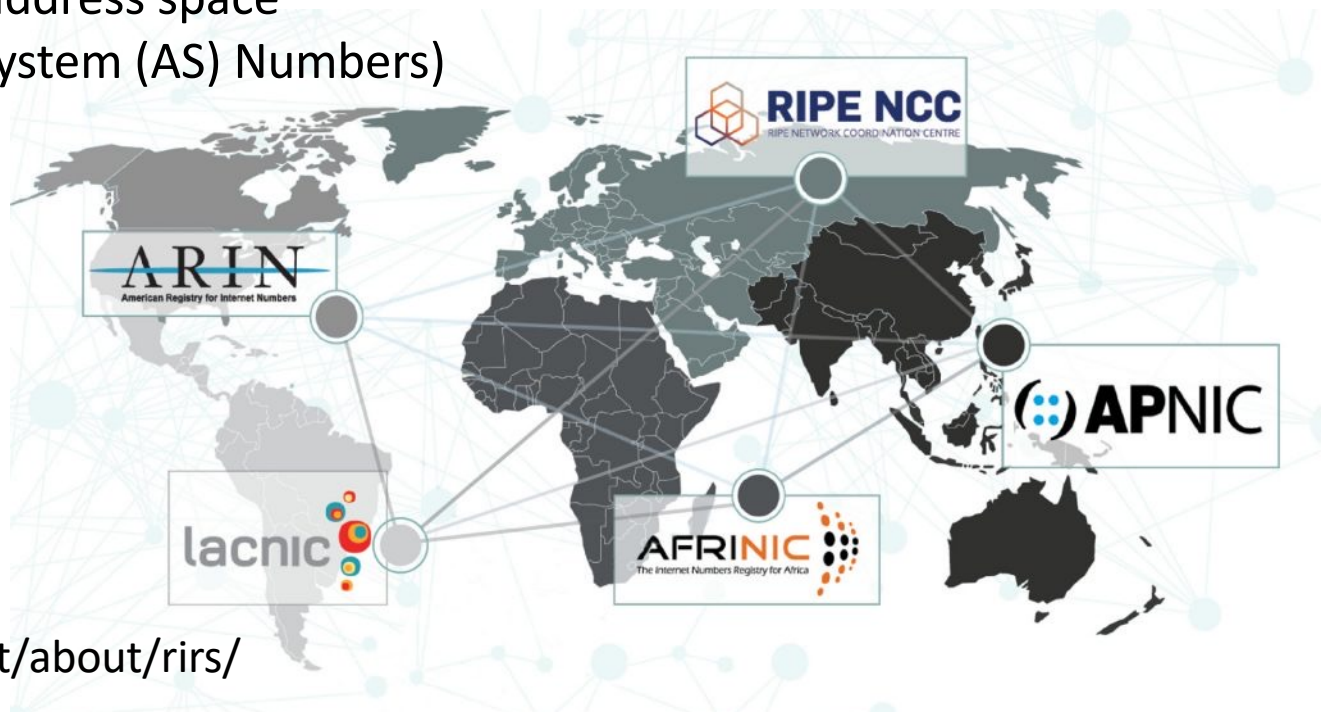
Internet Assigned Numbers Authority

“Affiliate” of ICANN

Regional Internet Registries (RIR)

RIRs manage, distribute, and register **Internet number resources** within their respective regions. Global policy (via ICANN) governs distribution to regions.

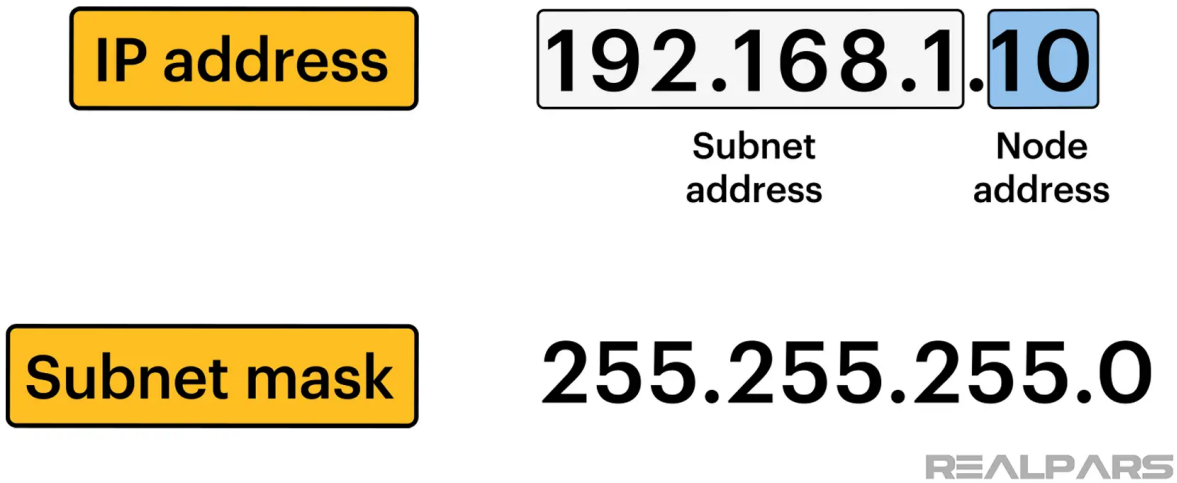
- IPv4 and IPv6 address space
- Autonomous System (AS) Numbers)



https://www.nro.net/about/rirs/

IPv4 dotted-quad notation

or dot-decimal

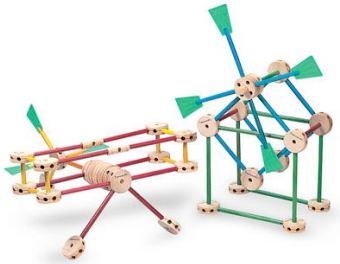


An RFC 1918 private subnet: 192.168 (11000000 101010)

IPv4 netmask

- In Internet Protocol IPv4 ("classic") each host/node has a 32-bit **IP address**.
- A "network" (or **subnet**) has a block of IP address space in which all addresses share the same **prefix**. An IP address prefix corresponds to a **network number**. The rest of the address bits are the **node number**.
- Prefixes are owned by entities that operate networks (e.g., Duke). Out on the Internet, routers route packets toward their destination address by looking up the prefix in a local routing table—like a zip code.
- How many bits in the prefix? Big networks need small prefixes. Small networks can have big prefixes. For example, a /24 subnet is small: it has a 24-bit network number and an 8-bit node number.
- A subnet's **netmask** is a constant to & with any IP address in the subnet to yield its network number: 0xFFFFFF00 for a /24, aka 255.255.255.0.

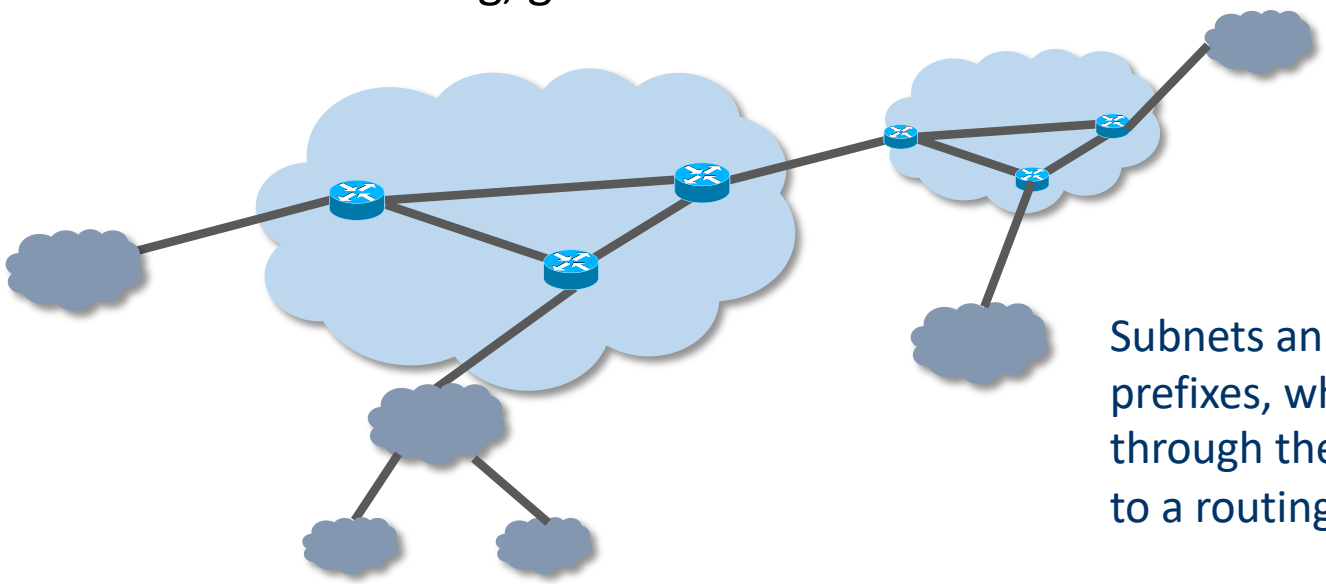
How to think about networks



- Build out networks by adding nodes and/or links.
- Stitch networks together into inter-networks.
  - Each network has one owner.
  - Inter-networks have multiple owners.
- A node may have multiple interfaces on different nets.
- Each **interface** (NIC device) has a name and/or number (an address) that is unique and distinct among all interfaces reachable from it through the links, transitively.
  - Examples: MAC address, IP address
- Networking topics: how addresses are assigned, structured, propagated, and stored; how packet routing works; transport protocols for reliable packet delivery.

The Internet (not to scale)

- A **node** or **host** is a computer attached to the net.
- Each host has a numeric **IP address** (at least one).
- Human-readable **domain names** map to IP addresses.
- **Issues**: secure naming, governance.



Subnets announce their IP prefixes, which propagate through the network according to a routing protocol.